Fake Currency Detection Using Deep Learning

1. Introduction

Fake Currency Detection Using Deep Learning is a data science project designed to identify whether a currency note is genuine or counterfeit by analyzing its image. Counterfeit currency can cause financial losses to individuals, businesses, and banks. Traditional methods of checking currency notes often require manual inspection and knowledge of security features.

This project uses Python, image processing, and deep learning techniques to analyze currency note images and classify them as real or fake. A Convolutional Neural Network (CNN) can be trained using a dataset containing images of genuine and counterfeit notes. After training, the model can predict the category of a new currency image.

The main objective of this project is to develop a simple, efficient, and user-friendly system that assists in currency verification. It can be useful for educational purposes and as a preliminary screening tool. However, the model's predictions should not be treated as definitive proof of authenticity.

2. Instructions for Using the Project

Follow these steps to develop and run the Fake Currency Detection system.



Step 1: Collect the dataset

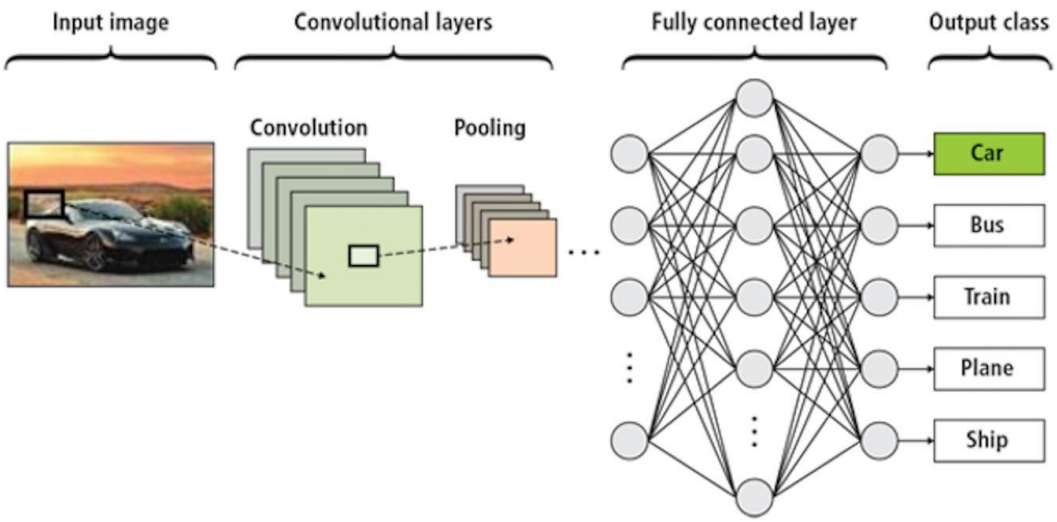
Collect images of genuine and counterfeit currency notes. Organize them into separate folders, such as Real and Fake. Use legally obtained images and ensure the dataset has reliable labels.



Step 2: Install the required libraries

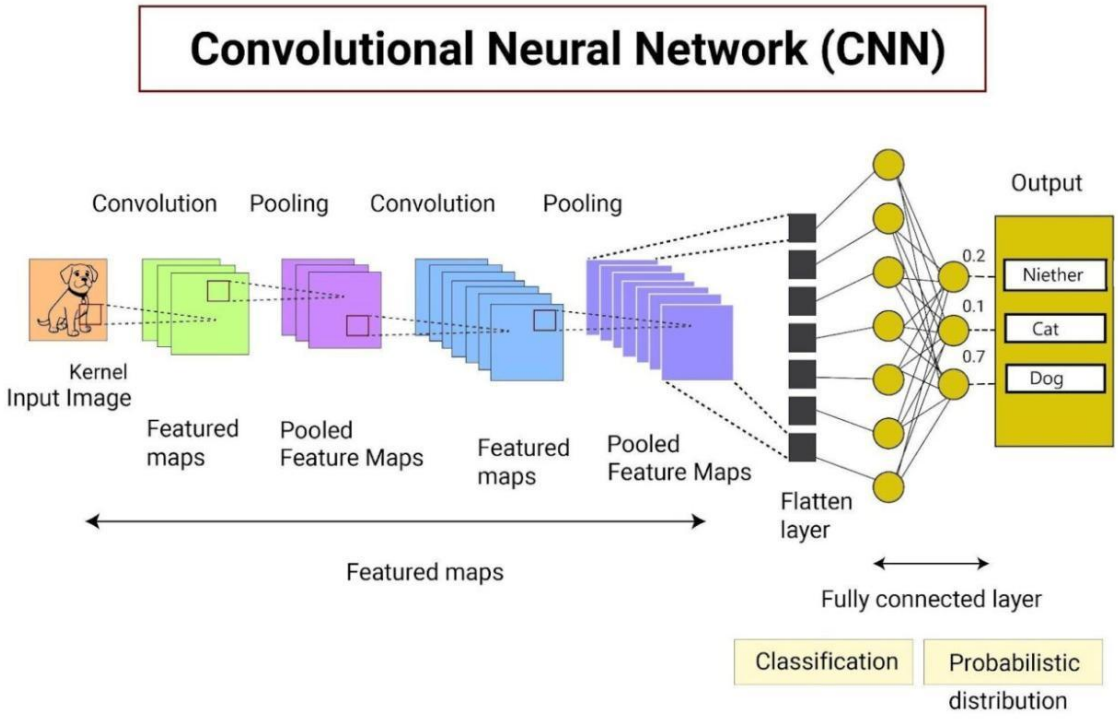
Install Python and the necessary libraries for data processing, image handling, visualization, and deep learning.

pip install numpy pandas matplotlib seaborn opencv-python tensorflow scikit-learn



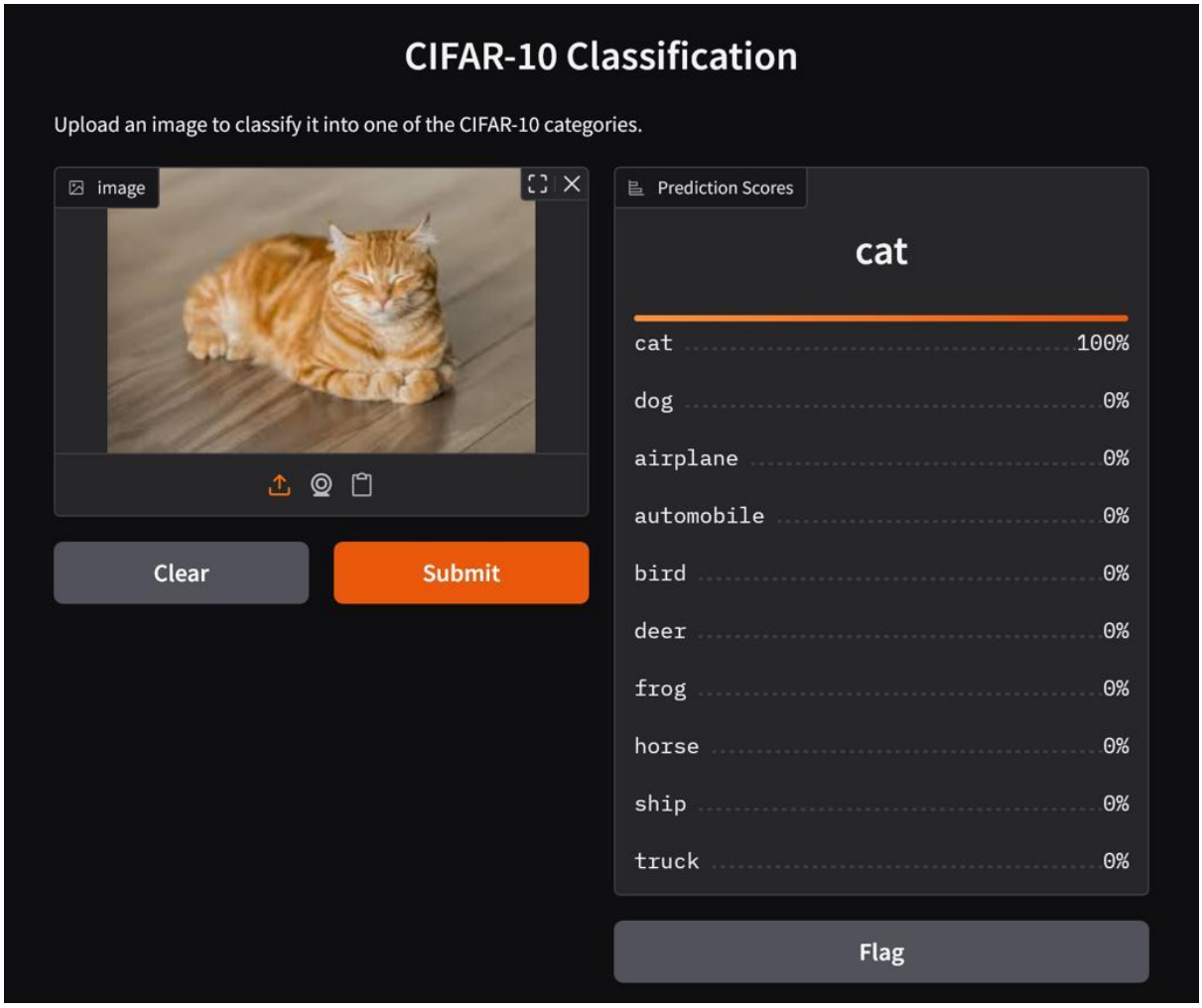
Step 3: Preprocess the images

Resize all images to a fixed size, normalize pixel values, and divide the dataset into training, validation, and testing sets.



Step 4: Train the deep learning model

Build a CNN model using TensorFlow/Keras. Train it on labeled images so it can learn visual patterns associated with the two classes.



Step 5: Test and evaluate the model

Evaluate the trained model on images it has not seen before. Measure accuracy, precision, recall, and the confusion matrix to understand its performance.



Step 6: Predict a new currency image

Upload a currency image to the application. The system processes the image and displays the model's predicted class: Real or Fake, along with an optional confidence score.

3. Software and Libraries Required

Software / Library	**Purpose**
Python	Main programming language
NumPy	Numerical operations
Pandas	Dataset management
Matplotlib and Seaborn	Data visualization
OpenCV	Image processing
TensorFlow/Keras	Building and training the CNN
Scikit-learn	Evaluation and data splitting
Jupyter Notebook or Google Colab	Writing and running the code

4. Expected Output

The system accepts a currency image and displays a prediction such as:

- Predicted class: Real currency

- Predicted class: Fake currency

These are example outputs, not actual verification results. Performance depends on the quality and diversity of the dataset. Image-based predictions alone cannot reliably establish authenticity; official security-feature checks or verification by an authorized institution may be necessary.

5. Important Instructions

1. Use a balanced dataset with correctly labeled images.
2. Avoid using the same or near-identical images in both training and testing.
3. Train and save the model before performing predictions.
4. Test the system under different lighting conditions, angles, and image qualities.
5. Do not rely on the model alone for financial or legal decisions.

Conclusion: Fake Currency Detection Using Deep Learning demonstrates how artificial intelligence and computer vision can assist in identifying potentially counterfeit banknotes. The project combines dataset collection, preprocessing, CNN training, model evaluation, and image prediction into one practical application.